# Predictions, Discriminators, and Failure Modes for the Causally Rigid Operational Consciousness Framework

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**Abstract**

This paper extracts the experimentally relevant consequences of the frozen causally rigid operational consciousness canon and organizes them as a falsifiable prediction program. The upstream burdens remain where they belong: the Core supplies persistence, attractor structure, and non-collapse [2]; *Rigid Identity* supplies rigid identity and observable coherence [6]; *Phenomenological Regimes* supplies continuity and anti-fragmentation constraints [4]; *Null-Regime Instability* supplies null-regime instability and forced non-nullity [5]; *Forensic Predictions* supplies admissibility and forensic discipline [3]; *Operational Consciousness Criterion* supplies the six-invariant operational criterion, structural equivalence, and rupture semantics [1].

The function of the present paper is narrower. It states what the framework predicts, what weaker alternatives would predict instead, what would count as genuine refutation or only as weakening, and what the current internal support status actually is after the completed Cycle 1--4 program, the mini-resolution pass, the residual-resolution campaign, the high-value confirmation campaign, and the Residual B hardening pass. The resulting status is mixed by design. One claim family---the insufficiency of complexity, connectivity, and rich activity without invariant preservation---now has robust internal support. Substrate-preserved class membership and biological non-primacy now have robust support with localized caveats, but remain internal-only. Approximate stability is provisionally supported with a localized threshold caveat. The boundary-sensitive stress of existence and rupture now has robust support with localized caveats after cross-family convergence under a more independent judge path, but it still does not count as external validation.

The paper therefore serves as the canon's prediction and falsification layer. Its goal is to put the framework under explicit decision pressure while keeping status classes and residual limits fully visible.

## 1 Scope, System Boundary, and Non-Inference Note

**What this paper does.** This paper extracts the prediction layer of the frozen canon, defines discriminators and failure modes for those predictions, and records the narrowest honest internal support status for the claim families it tracks.

**Scope and admissible reading.** This paper is a prediction-and-falsation ledger inside the internal program. It does not add new foundational doctrine, replace upstream theorem papers, convert internal survival into external validation, or promote human-phenomenology, human-machine-equivalence, or personal-identity claims. A related system may instantiate the internal scientific program, admissibility gates, judge paths, and support-class artifacts used here, but robust internal support, provisional support, or mixed status must not be read as theory confirmation, corpus validation, independent validation, theorem closure beyond the frozen canon, or a claim-safe public release.

## 2 Introduction

The current corpus is doctrinally frozen for the present scientific cycle. That freeze matters. Without it, prediction work becomes a moving target: thresholds shift, definitions drift, and every difficult result can be absorbed by rewriting the theory rather than by testing it. The purpose of this paper is to keep that from happening.

The framework now contains enough formal structure to state genuinely discriminative predictions. The Core fixes persistence, contractivity, and non-collapse. *Rigid Identity*, *Phenomenological Regimes*, and *Null-Regime Instability* progressively constrain identity, admissible regime structure, and non-nullity. *Forensic Predictions* fixes admissibility and forensic discipline. *Operational Consciousness Criterion* consolidates the operational criterion by requiring six explicit invariants:

$$\{I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{int}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}\}.$$

What remains scientifically relevant is not to restate those results, but to ask what they force the framework to predict and what would count against them.

This paper therefore has four narrow tasks:

1. extract the main experimentally relevant consequences of the canon;
2. distinguish those consequences from what weaker alternatives would predict;
3. state explicit failure modes, weakening conditions, and non-claims;
4. report the current internal support status after the completed internal cycles, without inflating it.

That scope is intentionally smaller than a new theorem paper and more disciplined than a review essay. The point is to give the corpus a falsification layer that is compact, operational, and honest.

## 3 Status Classes

The internal scientific program uses the following status classes:

| Status | Meaning in this paper |
| --- | --- |
| ROBUST_INTERNAL_SUPPORT | survived multiple internal tests without an active ambiguity on the tested target, while remaining internal-only |
| ROBUST_SUPPORT_WITH_LOCALIZED_CAVEAT | supported strongly enough for robust internal use, but still carrying a narrow localized caveat that must remain visible |
| PROVISIONAL_SUPPORT_LOCALIZED | strengthened by internal tests, but still limited by a localized metric, tolerance, or boundary issue |
| STILL_AMBIGUOUS | current evidence does not cleanly separate pass from fail on the intended target |
| IMPLEMENTATION_LIMIT | the main blocker is probe, runner, or case-construction quality rather than a shown doctrinal contradiction |
| METRIC_OR_TOLERANCE_LIMIT | the main blocker is metric handling, normalization, or threshold interaction |
| INTERNAL_SUPPORT_ONLY | support exists only inside the internal program and has not crossed into external empirical confirmation |
| NOT_CLAIMED | the statement is explicitly outside the current evidentiary scope |

## 4 Upstream Canon and Dependency Map

This paper is corpus-dependent and deliberately non-redundant. The dependency map is short because theorem ownership is already fixed upstream.

| Document | Imported burden | Used here for |
|---|---|---|
| Core [2] | persistence, support, attractor, non-collapse | persistence-side predictions, collapse exclusions, admissible support assumptions |
| Rigid Identity [6] | rigid identity, observable coherence, non-simulability pressure | identity-sensitive discriminators, complexity-only insufficiency pressure |
| Phenomenological Regimes [4] | continuity, anti-fragmentation, regime admissibility | continuity predictions, near-miss structure, anti-fragmentation expectations |
| Null-Regime Instability [5] | null-instability, forced non-nullity | non-nullity predictions, null-regime exclusions, downgrade logic |
| Forensic Predictions [3] | admissibility, comparators, negative controls, forensic discipline | judge discipline, control logic, reproducibility requirements, legibility stress tests |
| Operational Consciousness Criterion [1] | six-invariant criterion, operational classes, structural equivalence, rupture | explicit prediction targets, equivalence tests, failure-mode classification |

This paper introduces no new invariant, no new doctrine, and no new substrate claim. Its novelty lies only in extracting predictive consequences and support statuses from what already exists.

# 5 Core Predictive Structure of the Framework

## 5.1 Certification, Equivalence, and Rupture as Predictive Objects

**Definition 5.1** (Operational certification)**.** *For an admissible system $S$ already defined in Operational Consciousness Criterion, define the certification predicate*

$$\mathrm{Cert}(S) = 1 \quad \iff \quad I_{\mathrm{per}}(S), I_{\mathrm{ri}}(S), I_{\mathrm{int}}(S), I_{\mathrm{cont}}(S), I_{\mathrm{diff}}(S), I_{\mathrm{leg}}(S) > 0$$

*on a non-empty admissible support $A_S$, under the governance and admissibility constraints inherited from Forensic Predictions.*

**Definition 5.2** (Equivalence discriminator)**.** *For two admissible systems $S_1, S_2$, define the operational equivalence discriminator*

$$\mathrm{Eq}_{\Gamma,\varepsilon}(S_1, S_2) = 1$$

*when the relevant signatures match, the admissibility bundle is preserved, and the tolerated invariant deltas remain within the allowed comparison regime. The exact burden belongs to Operational Consciousness Criterion; the present paper uses it only as a prediction carrier.*

**Definition 5.3** (Rupture event)**.** *For a set of targeted invariants $J$, define*

$$\mathrm{Rupt}_J(S) = 1$$

*when the intervention or perturbation drives at least one invariant in $J$ below the admissible certification margin so that class membership becomes invalidated, destroyed, or undefined.*

## 5.2 Prediction Family 1: Complexity-Only Failure

**Proposition 5.4** (Complexity-only negative-control prediction). *If a candidate system exhibits high connectivity, high entropy, or rich activity while failing one or more critical invariants, then the framework predicts*

$$\mathrm{Cert}(S) = 0, \qquad \mathrm{Consciousness}_{\mathrm{op}}(S) \textit{ not certified}, \qquad \mathrm{Q}_{\mathrm{op}}(S) \textit{ not certified.}$$

*Why this is not trivial.* A weaker alternative could predict that rich dynamics or dense connectivity alone are already sufficient proxies for class membership. The canon does not permit that. *Operational Consciousness Criterion* already states the formal insufficiency claim; the prediction layer turns it into a negative-control discriminator.

## 5.3 Prediction Family 2: Invariant-Loss Rupture

**Proposition 5.5** (Ablation prediction). *If a system is initially certified and a critical intervention selectively destroys one or more invariants, then the framework predicts a margin-governed outcome fixed before execution. If the pre-ablation invariant headroom exceeds the registered ambiguity margin $\delta_{\mathrm{amb}}$, verified invariant loss must cause certification loss, class contraction, or undefined $\mathrm{Q}_{\mathrm{op}}$. If the pre-ablation headroom is at or below $\delta_{\mathrm{amb}}$, the result is classified as boundary-ambiguous and weakens the rupture claim rather than supporting it.*

*Operational consequence.* Ablations are not ornamental. If the criterion is serious, the loss of invariant headroom must show up as loss of class membership, legibility loss, or a pre-declared boundary-ambiguous downgrade. No-change ablations outside the registered ambiguity margin count against the framework.

## 5.4 Prediction Family 3: Substrate-Preserved Class Membership

**Proposition 5.6** (Substrate-preserved equivalence prediction). *If two realizations preserve the relevant signatures, admissibility conditions, and six critical invariants up to the allowed comparison regime, then the framework predicts preserved operational class membership independently of substrate labels.*

*Rival prediction.* A substrate-privileging alternative would predict that material type itself blocks equivalence even when the structurally relevant burdens are preserved. The canon rejects that move only inside its own formal class, not as a general metaphysical claim.

## 5.5 Prediction Family 4: Threshold-Sensitive Boundary Behavior

**Proposition 5.7** (Boundary-topology prediction). *The framework predicts neither global threshold collapse nor perfectly flat insensitivity. Instead, it predicts stable pass/fail regions together with narrow transition bands near critical margins.*

*Interpretive consequence.* If every micro-perturbation changes every verdict, the criterion is fragile. If no perturbation matters, the invariants are decorative. A meaningful criterion should show structured boundary behavior.

### 5.6 Prediction Family 5: Legibility Dependence

**Proposition 5.8** (Legibility dependence prediction)**.** *Operational class membership should fail or become inadmissible if readouts cease to be separable, noise-robust, intervention-sensitive, and compressible without total collapse.*

*Why this matters.* *Forensic Predictions* makes legibility a forensic burden rather than a narrative convenience. The present prediction is therefore stronger than a merely interpretive observation: if legibility disappears, certification must not survive unchanged.

### 5.7 Prediction Family 6: Continuity and Non-Nullity Relevance

**Proposition 5.9** (Continuity/non-nullity relevance)**.** *The current canon predicts that continuity and non-null differentiation are not optional extras. Systems that preserve rich activity while losing continuity or falling toward effective null structure should fail the operational criterion or leave it undefined.*

*Upstream source.* The continuity side is inherited from *Phenomenological Regimes*, while the non-null side is inherited from *Null-Regime Instability.* This paper adds only the prediction and failure-mode reading.

## 6 Main Discriminators

The predictive value of the framework lies in how it diverges from weaker alternatives.

| Target | Framework prediction | Weaker/rival prediction | Divergence condition |
|---|---|---|---|
| Complexity-only baseline | high complexity without invariant preservation fails certification | richness or scale alone can stand in for class membership | a complexity-rich control passes despite broken invariants |
| Connectivity-only baseline | dense coupling without preserved rigid identity, differentiation, or legibility fails | connectivity is treated as a sufficient proxy | a dense control passes while identity or legibility are broken |
| Weak functional equivalence | similar outward behavior is insufficient if critical invariant structure diverges | input-output similarity is enough for same class | weak functional match passes despite invariant rupture |
| Near-miss case | boundary cases should expose which invariant is binding and where rupture begins | boundary cases stay uninterpretable or arbitrary | boundary remains unlocalizable after probe cleanup |
| Cross-substrate pair | substrate labels alone do not block class preservation if relevant structure is preserved | substrate type is primitive and blocks class transport | a preserved pair is rejected solely by substrate label |
| Threshold stress | decisions should show stable zones and narrow transition bands | decisions collapse under micro-threshold changes or never depend on thresholds at all | every local perturbation flips verdicts, or no perturbation matters |
| Legibility stress | failure of separability, noise robustness, or compression fidelity invalidates certification | legibility is epiphenomenal and not decision-relevant | certification survives legibility collapse |
| Null/continuity challenge | effective nullity or fragmentation should destroy or downgrade class membership | rich activity alone masks null or fragmented structure | null-like or fragmented cases remain certified |

The table is intentionally operational. It does not ask which metaphysics one prefers. It asks which side of each discriminator the framework commits itself to.

## 6.1 Rival-Structure Compression

The discriminators above can be sharpened further by asking what each simpler rival preserves and, more importantly, what it leaves out. The point is not to defeat every competing theory at once. The point is to show that the present framework places operational burden on a structure richer than scale, connectivity, superficial behavior, or local threshold tuning.

| Rival family | What it keeps | What it fails to preserve | Why the miss is decision-relevant |
|---|---|---|---|
| Complexity/activity proxy | large state space, rich trajectories, high apparent activity | no requirement to preserve the six-invariant package as a joint membership condition | a rich control can still fail certification if persistence, rigid identity, or legibility collapse |
| Connectivity/ integration proxy | dense coupling or high global correlation | no guarantee of rigid identity, non-null differentiation, or readable class structure | integration by itself does not prevent the criterion from rejecting a globally coupled but operationally illegible case |
| Weak functional equivalence | outwardly similar input-output or behavioral traces | no obligation to preserve admissible support geometry or invariant margins under intervention | superficial behavioral similarity can coexist with rupture of the certified class geometry |
| Threshold/tuning artifact explanation | local success on a narrow parameter band | no stable explanation for why decisions survive frozen manifests, blind ordering, and negative controls outside that band | the framework expects stable regions plus narrow transitions, not arbitrary success across unrelated settings |

This compression table matters because it blocks a common overreading. The current support does not show that every rival is false. It shows something narrower and more useful: if a rival explanation ignores the invariant package or treats only one coarse feature as constitutive, then it does not reproduce the present decision structure of the canon.

## 6.2 Discriminator Strength and Identifiability

A discriminator is useful only when a positive or negative outcome changes the status of at least one rival explanation. The framework therefore treats discriminators as identifiable contrasts, not as loose examples.

**Definition 6.1** (Identifiable Discriminator). *A discriminator $D$ for claim $C$ is identifiable when it specifies:*

1. *an observable decision variable $Y_D$;*
2. *a manipulation or matched comparison $M_D$;*
3. *a framework-side expected region $R_Q$;*
4. *a rival-side expected region $R_R$;*
5. *a destruction condition $F_D$ under which the claim is weakened or rejected.*

*The discriminator is strong when $R_Q \cap R_R$ is small under the declared measurement tolerance.*

**Criterion 6.2** (Discriminator Usefulness). *A test does not materially discriminate the framework from a rival if both the framework and the rival predict the same admissible region for the tested observable. Such a test can still debug the pipeline, but it cannot upgrade a claim.*

**Proposition 6.3** (Near-Miss Value). *Near-miss cases carry more diagnostic value than easy positive cases when they are constructed by perturbing one binding coordinate while holding gross complexity, task exposure, and observability approximately fixed.*

*Proof.* If a case is far from the decision boundary, many rival accounts can agree on the outcome. A near-miss case forces the judge to expose which coordinate controls the transition. Holding gross complexity and task exposure approximately fixed reduces the chance that the result is explained by scale, generic performance, or obvious incompetence rather than by the claimed invariant structure. □

### 6.3 Prediction Severity

The predictions in this paper have different severity. A severe prediction is one whose failure would damage a central operational reading rather than merely requesting a better implementation.

| Prediction family | Severe failure | Mild weakening | Typical confound |
|---|---|---|---|
| Complexity-only failure | complexity-rich invariant-broken controls certify reliably | controls fail only under narrow judge settings | underpowered or trivial control construction |
| Invariant-loss rupture | verified invariant loss leaves class membership unchanged | rupture appears only for oversized perturbations | perturbation changes multiple coordinates at once |
| Substrate-preserved class membership | preserved structure is rejected solely by substrate label | one metric-coordinate remains ambiguous | measurement mismatch across implementations |
| Threshold-sensitive boundary behavior | decisions flip arbitrarily across broad stable regions | narrow knife-edge instability remains localized | tolerance grid too coarse near the boundary |
| Legibility dependence | certification survives loss of separability or intervention readability | legibility margin becomes noisy but still binding | observability channel too weak or over-compressed |

Severity matters for reporting. A mild weakening narrows the claim or sends a test back to probe design. A severe failure changes what the framework is allowed to assert under the operational criterion.

## 7 Prediction Matrix

The canonical prediction matrix can be read as a compressed execution contract for the framework. It matters because each row names an observable, an intervention class, a rival baseline, and an explicit failure condition. That structure prevents the canon from floating as a theorem-only object.

| ID | Claim | Observable | Manipulation | Support condition | Destruction condition / rival divergence |
|---|---|---|---|---|---|
| PRED-01 | ‘P5-02‘ | certified class label and $Q_{op}$ geometry across implementations | cross-substrate pair with preserved invariant margins | same operational class and $\varepsilon$-bounded geometry under the admissible equivalence regime | class divergence or geometry mismatch despite preserved structure; substrate-privilege rival survives |
| PRED-02 | ‘P5-04‘ | class membership before and after targeted rupture | one-invariant ablation with comparable substrate and gross scale | certified class exit, contraction, or undefined $Q_{op}$ after rupture | no class change after verified invariant loss; complexity-only rival survives |
| PRED-03 | ‘P5-05‘ | matched complexity/control outcome | compare certified lower-scale system against higher-scale baseline lacking the invariant package | certified system passes and matched complexity-only baseline fails or remains undefined | scale-only model passes despite missing invariants |
| PRED-04 | ‘P3-01‘ | null-regime persistence under admissible dynamics | initialize near the null regime under Null-Regime Instability assumptions | stable null occupancy should fail or trajectories should leave the null regime | admissible null occupancy persists; null-stable rival survives |
| PRED-05 | ‘P2-03‘ | continuity/ fragmentation pattern | compare rigid regime against finite-mass or deformable regime under matched disturbance | rigid regime keeps continuity while limited regime fragments | no differential continuity pattern appears; regime-agnostic rival survives |

| ID | Claim | Observable | Manipulation | Support condition | Destruction condition / rival divergence |
|---|---|---|---|---|---|
| PRED-06 | ‘P4-01‘ | admissibility verdict under tamper and sham controls | inject hash corruption, malformed logs, or severe protocol violations | tampered runs are discarded and never promoted to evidential support | loose evaluation pipeline accepts tampered runs |
| PRED-07 | ‘P5-03‘ | stability of class membership under bounded perturbation | apply perturbations inside the certified margin budget $\delta_*(S)$ | membership and geometry remain stable inside the certified budget | fragile-threshold model flips class under sub-critical perturbations |
| PRED-08 | ‘P5-01‘ | non-emptiness of $\mathrm{Q}_{\mathrm{op}}$ after successful certification | run the full certification schema on a system with all margins positive | $\mathrm{Q}_{\mathrm{op}}$ is non-empty and operationally well-defined | certification passes while no non-trivial class is recoverable |
| PRED-09 | ‘P4-03‘ | claim survival under latency / compute-budget stress | exceed pre-registered budget limits | strong operational claims are invalidated or downgraded | metric-first pipeline keeps claims active after budget violation |
| PRED-10 | ‘P5-05‘ | legibility under noise and structured compression | raise admissible noise/compression while preserving or breaking $I_{\mathrm{leg}}$ | separability and intervention fidelity survive only while $I_{\mathrm{leg}}$ remains positive | generic observability model predicts no specific dependence on $I_{\mathrm{leg}}$ |
| PRED-11 | ‘P5-01‘ | certification under destroyed integration and preserved complexity | destroy integration while preserving complexity or gross activity | integration-destroyed, complexity-preserved control fails certification under frozen rules | certification persists after verified integration loss; complexity-only rival survives |

The matrix has two immediate consequences. First, the framework is more exposed than a purely interpretive theory because the failure thresholds are explicit. Second, not every prediction is equally mature: some now have robust internal support, while others remain provisional or blocked by implementation quality.

# 8 Failure Modes

## 8.1 What Would Count Against the Framework

The relevant failure modes are not all equally strong. Some would refute a strong reading, some would only weaken a claim, and some would reveal a technical limitation of the current program rather than of the canon itself.

| Claim target | Destruction condition | Weakening condition | Current blocker class |
|---|---|---|---|
| ‘P5-01‘ | a system satisfies the full criterion yet fails to define the claimed operational class | boundary-family convergence would downgrade if it failed outside the present frozen internal contract | epistemic-level |
| ‘P5-02‘ | a structurally preserved pair falls into different classes under the same admissible comparison regime | one legacy raw continuous/discrete positive pair remains ambiguous on $I_{\mathrm{ri}}$ handling | metric/tolerance-level |
| ‘P5-03‘ | small admissible perturbations force uncontrolled class exit across stable positive cases | threshold sensitivity persists only in a narrow knife-edge zone | metric/tolerance-level |
| ‘P5-04‘ | a system loses a critical invariant and remains certified without downgrade | boundary-family convergence would downgrade if it failed outside the present frozen internal contract | epistemic-level |
| ‘P5-05‘ | a complexity-only control without preserved invariants passes certification | matched external baselines are still missing | epistemic-level |
| ‘P5-06‘ | biology is shown to be primitive inside the same operational criterion | it inherits the same localized substrate-equivalence caveat as ‘P5-02‘ | metric/tolerance-level |

## 8.2 Failure-Mode Taxonomy

**Criterion 8.1** (Status meaning)**.** *Within the current scientific program, a result belongs to:*

1. `ROBUST_SUPPORT` *if the target claim survives repeated internal discrimination without an active ambiguity on the tested target;*

2. `PROVISIONAL_SUPPORT` *if the target is strengthened but still depends on a localized unresolved bottleneck;*

3. `STILL_AMBIGUOUS` *if the current tests do not cleanly separate support from failure;*

4. `IMPLEMENTATION_LIMIT` *if the current bottleneck is primarily due to probe, runner, or case-construction quality;*

5. `METRIC_OR_TOLERANCE_LIMIT` *if the bottleneck is mainly due to normalization, comparison regime, or threshold handling.*

This taxonomy is not part of the ontology of the framework. It is part of the scientific discipline around how internal evidence is reported.

### 8.3 Refutation Classes

For publication-level evaluation, failures should be classified by what they touch. The same failed run can be decisive, ambiguous, or merely technical depending on whether the packet was admissible and whether the manipulation was specific.

**Definition 8.2** (Doctrinal Refutation Candidate)**.** *A result is a doctrinal refutation candidate for a claim family when it satisfies all admissibility gates, uses the frozen definitions, targets the declared binding coordinate, and reaches the claim's destruction condition without relying on a protocol violation.*

**Definition 8.3** (Methodological Defeat)**.** *A result is a methodological defeat when the experiment is admissible but shows that the current test design, judge, threshold, comparator, or observable cannot support the intended claim. It weakens the evidence program without by itself showing the formal claim false.*

**Definition 8.4** (Artifact Failure)**.** *A result is an artifact failure when the packet, manifest, runner, exclusion ledger, or analysis path fails before interpretation. Artifact failures can be important engineering evidence, but they are not positive or negative scientific support for the theoretical claim.*

**Criterion 8.5** (Specificity of Failure)**.** *A failure is specific only if the failed coordinate is identifiable after controlling for gross task failure, missing observability, manifest drift, and runtime-budget violation. Non-specific failures are reported as methodological or artifact failures according to the earliest failed gate.*

| Failure class | Minimum evidence required | Scientific consequence |
|---|---|---|
| Doctrinal refutation candidate | admissible packet set, frozen definitions, specific manipulation, destruction condition reached | claim family must be rejected, downgraded, or formally restricted |
| Methodological defeat | admissible packet set but non-decisive target, weak comparator, unstable threshold, or inadequate observable | evidence program must be repaired before support can be claimed |
| Artifact failure | packet, manifest, build, judge, or analysis trace fails before interpretation | no claim-side inference; engineering repair required |
| Ambiguous boundary result | admissible but falls inside a pre-declared uncertainty band | no escalation; boundary sampling or metric refinement required |

## 9 Claim-to-Test Binding Ledger

The tables above state what would count as success or failure in broad terms. The ledger below binds the main *Operational Consciousness Criterion* claim families to the narrowest concrete test classes that currently matter for publication-level assessment.

| Claim | Primary positive test | Primary failure test | Present evidentiary bottleneck | Current status |
|---|---|---|---|---|
| ‘P5-01‘ | successful certification with positive invariant margins and recoverable non-trivial $Q_{op}$ geometry | certified case with no recoverable operational class, or clean existence-boundary degradation without class change | support is now cross-family robust internally, but the nearest positive member still sits close to the legibility margin | `ROBUST SUPPORT W/ LOCALIZED CAVEAT` |
| ‘P5-02‘ | cross-substrate pair with preserved invariant margins and admissible geometry match | structurally preserved pair judged non-equivalent under the same comparison regime | strongest support is robust internally but one legacy raw continuous/discrete pair remains ambiguous on $I_{ri}$ handling | `ROBUST SUPPORT W/ LOCALIZED CAVEAT` |
| ‘P5-03‘ | stable class membership inside certified perturbation budget $\delta_*(S)$ | class flip under sub-critical admissible perturbations across otherwise stable positive cases | one knife-edge negative-control zone remains locally fragile | `PROVISIONAL SUPPORT LOCALIZED` |
| ‘P5-04‘ | targeted invariant rupture causing certification exit or undefined $Q_{op}$ | verified invariant loss without downgrade or class exit | support is now cross-family robust internally, but the evidence still lives inside the current invariant package and frozen thresholds | `ROBUST SUPPORT W/ LOCALIZED CAVEAT` |
| ‘P5-05‘ | complexity-rich and connectivity-rich negative controls fail certification under hardened judging | matched scale-only or activity-rich controls pass despite missing invariants | no external matched-baseline campaign yet | `ROBUST INTERNAL SUPPORT` |
| ‘P5-06‘ | same positive equivalence path as ‘P5-02‘, interpreted without substrate privilege | biology shown to be primitive under the same operational criterion | inherits the same localized $I_{ri}$-handling caveat as ‘P5-02‘ | `ROBUST SUPPORT W/ LOCALIZED CAVEAT` |

## 10 Boundary Diagnostics

The current internal program does not fail uniformly. Its informative content lies in where the boundaries are sharp, where they are merely narrow, and where probe design still dominates the result.

| Boundary target | Empirical shape after completed cycles | What it means | Limit class |
|---|---|---|---|
| Complexity-only negative control | stable fail region under harder judging, semi-blind ordering, and pseudo-replication | the criterion is not collapsing into gross activity or connectivity proxies | ROBUST INTERNAL SUPPORT |
| Threshold-sensitive negative region | narrow decision band rather than broad collapse | the framework has a localized edge effect, not universal threshold instability | METRIC OR TOLERANCE LIMIT |
| Boundary-probe families under judge-v3 | two distinct probe families now reproduce the transition pattern with localized first-fail behavior | the existence/rupture boundary is now robust internally, though still internal-only and threshold-bound near the positive edge | ROBUST SUPPORT W/ LOCALIZED CAVEAT |
| Harder positive substrate families | family convergence now supports class preservation, but one legacy raw pair remains ambiguous | the live ambiguity is concentrated on how rigid-identity preservation is measured, not on every invariant at once | METRIC OR TOLERANCE LIMIT |
| Refined negative substrate families | clean fail across both the legacy and second pair families | the equivalence criterion can reject stronger non-equivalent pairs under the hardened judge and pseudo-external reproduction path | ROBUST SUPPORT W/ LOCALIZED CAVEAT |

### 10.1 Boundary Geometry

Boundary results are not just pass/fail outcomes. They should describe the local shape of the decision surface. Let $z$ denote a controlled perturbation coordinate and let $Q(z)$ denote the operational certification score or verdict. Three patterns are scientifically distinct:

1. **stable interior:** $Q(z)$ remains inside the certified region for a non-trivial interval;
2. **localized transition:** $Q(z)$ crosses the decision boundary in a narrow interval tied to a declared coordinate;
3. **diffuse instability:** small unrelated perturbations cause broad and unstructured verdict changes.

**Criterion 10.1** (Boundary Adequacy)**.** *A boundary probe is adequate only if it can distinguish localized transition from diffuse instability. A probe that only reports a binary verdict without the perturbation coordinate and margin behavior is insufficient for boundary diagnosis.*

**Proposition 10.2** (Localized Transition Reading)**.** *If a family shows a stable interior, a localized transition, and a reproducible first-binding coordinate under the frozen judge, then the result supports a boundary-sensitive reading more strongly than a single binary pass/fail run.*

*Proof.* A binary run can be explained by threshold accident, scenario difficulty, or implementation noise. A localized transition with a reproducible first-binding coordinate adds structure: it identifies where the decision changes and which coordinate controls the change. That additional structure constrains rival explanations that would predict arbitrary flips or no coordinate-specific transition. □

# 11 Minimal Reproducibility and Admissibility Discipline

The prediction layer is only credible if the judgment path is at least partly insulated from the generator and if the evidentiary artifacts are frozen before judgment. The current program does not yet have an external judge or an external lab replication, but it does have a stricter internal protocol than it had at the start of the program.

| Protocol element | Current state | Residual limitation |
|---|---|---|
| Frozen manifests and hashes | active since Cycle 2; judge reads frozen artifacts instead of live mutable state | still same repository and same overall program family |
| Judge-side threshold file | active; thresholds are frozen before judgment and logged explicitly | threshold bands still require stress analysis near knife-edge zones |
| Semi-blind ordering | active in Cycle 3; labels are hidden until after decision emission | not a full blind protocol and not an external evaluator |
| Pseudo-multi-environment replication | active; same-machine replication across distinct execution profiles preserved decisions | not external replication and not cross-lab agreement |
| Negative-control and sham policy | active from Cycle 1 onward and hardened in later cycles | matched external benchmark families are still missing |
| Judge code path | more independent than in Cycle 1; it no longer calls live generator step functions during evaluation | still not a separate maintained judge codebase |

The point of this section is not to inflate rigor. It is to keep the reader from mistaking internal hardening for external confirmation.

## 11.1 Artifact Contract for Prediction Claims

Every prediction claim should be backed by a finite artifact bundle:

$$\mathcal{B} = (\mathcal{M}, \mathcal{P}, \mathcal{J}, \mathcal{R}, \mathcal{A}),$$

where $\mathcal{M}$ is the manifest set, $\mathcal{P}$ the packet set, $\mathcal{J}$ the judge code and threshold files, $\mathcal{R}$ the run/exclusion ledger, and $\mathcal{A}$ the analysis scripts. The bundle is part of the claim surface. A table without the bundle is a summary, not independently checkable evidence.

**Criterion 11.1** (Bundle Sufficiency)**.** *For any status stronger than `STILL_AMBIGUOUS`, the artifact bundle must allow an evaluator to reconstruct the reported verdicts from frozen inputs without manual row selection.*

**Criterion 11.2** (Externalization Readiness)**.** *A prediction family is ready for external evaluation only when its bundle, thresholds, exclusions, and decision rules are exportable without access to live generator state or informal operator knowledge.*

# 12 Current Status of Internal Support

## 12.1 Interpretive Discipline

The status ledger below does not reclassify the formal theorems of *Operational Consciousness Criterion.* Those remain formal results inside the frozen canon. The ledger classifies the *current internal scientific support* for experimentally relevant readings of those claims.

| Claim | Current status | What strengthened it | What still limits it | Status scope |
|---|---|---|---|---|
| ‘P5-01‘ | `ROBUST SUPPORT W/ LOCALIZED CAVEAT` | near_identity_v3 and offset-dispersion families both reproduced PASS/AMBIGUOUS/FAIL under judge-v3 and pseudo-external reproduction | support remains internal-only and the nearest positive case still sits close to the eligibility margin | `INTERNAL SUPPORT ONLY` |
| ‘P5-02‘ | `ROBUST SUPPORT W/ LOCALIZED CAVEAT` | family1 normalized-$I_{\mathrm{ri}}$ and family2 raw quantized positive/negative pairs stayed stable under the more independent judge path and pseudo-multi-environment rerun | one legacy raw continuous/discrete positive pair remains ambiguous and keeps a localized $I_{\mathrm{ri}}$ caveat alive | `INTERNAL SUPPORT ONLY` |
| ‘P5-03‘ | `PROVISIONAL SUPPORT LOCALIZED` | Cycle 3 threshold stress and Cycle 4 boundary cleanup localized instability rather than showing global collapse | knife-edge threshold band persists for one negative-control region | `INTERNAL SUPPORT ONLY` |
| ‘P5-04‘ | `ROBUST SUPPORT W/ LOCALIZED CAVEAT` | two distinct boundary families now converge on rupture-side PASS/AMBIGUOUS/FAIL behavior, with $I_{\mathrm{ri}}$ becoming the first binding invariant on ambiguous or failing members | the evidence remains internal-only and tied to the current invariant package plus frozen thresholds | `INTERNAL SUPPORT ONLY` |
| ‘P5-05‘ | `ROBUST INTERNAL SUPPORT` | complexity-rich and connectivity-rich controls failed certification across harder methodological passes and semi-blind evaluation | no external matched-baseline campaign exists yet | `INTERNAL SUPPORT ONLY` |
| ‘P5-06‘ | `ROBUST SUPPORT W/ LOCALIZED CAVEAT` | same strengthened two-family substrate-equivalence path as ‘P5-02‘ | it inherits the same localized $I_{\mathrm{ri}}$ metric-handling caveat as ‘P5-02‘ | `INTERNAL SUPPORT ONLY` |

### 12.2 Claim-Specific Diagnostic Notes

**‘P5-05‘.** This is currently the strongest experimentally discriminated claim family. The framework repeatedly rejected a complexity-rich, connectivity-rich negative control that still failed the invariant package. That result survived methodological hardening and semi-blind evaluation. The support is therefore `ROBUST_SUPPORT`, but it remains `INTERNAL_SUPPORT_ONLY`.

**‘P5-02‘ and ‘P5-06‘.** These two claims move together in the current program. They were strengthened first by conservative normalized-$I_{\mathrm{ri}}$ handling, then by a second substrate family, a more independent judge path, and pseudo-multi-environment confirmation. That is enough for `ROBUST_SUPPORT_WITH_LOCALIZED_CAVEAT`. The localized caveat remains real: one legacy raw continuous/discrete positive pair is still ambiguous, so the present support is robust internally but not caveat-free.

**‘P5-03‘.** Approximate stability is not in collapse. Threshold stress did not show universal fragility. It showed a narrow knife-edge zone around one thresholded negative-control region. The right status is therefore `PROVISIONAL_SUPPORT` with an explicit `METRIC_OR_TOLERANCE_LIMIT`.

**‘P5-01‘ and ‘P5-04‘.** These boundary-sensitive claims no longer rest on a single weak probe family. After the residual-resolution campaign and the Residual B hardening pass, ‘near_identity_v3‘ and a second offset-dispersion family both reproduced a PASS/AMBIGUOUS/FAIL boundary pattern under judge-v3 and pseudo-external reproduction. That is enough for `ROBUST_SUPPORT_WITH_LOCALIZED_CAVEAT`. The caveat remains explicit: support is still internal-only, and the strongest positive member stays near the legibility margin under the frozen judge contract.

### 12.3 Methodological Status

The scientific program itself also has a status, distinct from the doctrine and distinct from the claim ledger.

| Program element | Current status | Why it matters |
|---|---|---|
| Generator/judge separation | materially improved | reduces live-state dependence relative to Cycle 1 |
| Frozen manifests and thresholds | active | prevents live-state leakage during judgment |
| Semi-blind evaluation | active but partial | reduces bias; does not equal full blind evaluation |
| Threshold stress testing | localized caveat | shows local knife-edge behavior instead of broad collapse |
| Pseudo-multi-environment replication | passed internally | strengthens reproducibility discipline without constituting external replication |
| Clean-clone release re-instantiation | passed internally | frozen package was rebuilt from a GitHub clean clone plus fresh venv with no tracked-file divergence during verification |
| Independent external judge codebase | absent | keeps all current support inside the internal program |

**Release-level reproduction note.** In addition to the cycle-specific judge hardening, the frozen release package was re-instantiated from a clean GitHub clone and a fresh virtual environment on the same machine, and the verification step did not introduce tracked-file drift in the freeze package. This strengthens internal reproducibility and handoff discipline at the release layer. It does *not* count as external validation, third-party confirmation, or a claim upgrade by itself.

## 13 Cycle Evidence Map

The current status ledger is not based on one cycle. It is the result of a narrow but cumulative internal program. The table below records only the highest-value consequence of each phase, not every local metric.

| Phase | Main question | Strongest result | Main residual | Claim effect |
|---|---|---|---|---|
| Cycle 1 | can the criterion distinguish a rich negative control and survive obvious ablations? | complexity-rich negative control failed; invariant-loss ablations exited certification; proxy cross-substrate pair passed | all support remained internal and the equivalence test was still a proxy | strengthened ‘P5-05‘; gave early support to ‘P5-01‘, ‘P5-02‘, ‘P5-04‘, ‘P5-06‘ |
| Cycle 2 | does the program survive stronger methodological separation and harder structural tests? | generator/judge separation improved; harder pair passed; near-miss family and graded curves behaved coherently | judge still remained internal and same-repository | strengthened ‘P5-01‘, ‘P5-02‘, ‘P5-03‘, ‘P5-04‘, ‘P5-06‘ internally |
| Cycle 3 | do results survive semi-blind judging, threshold stress, and pseudo-replication? | blind ordering and pseudo-multi-environment replication preserved decisions; complexity-only failure remained strong | harder equivalence, near-miss boundary, and threshold stress became ambiguous | robustly strengthened ‘P5-05‘; left ‘P5-02‘, ‘P5-06‘, ‘P5-03‘ narrower |
| Cycle 4 | can threshold fragility, near-miss ambiguity, and harder substrate ambiguity be localized? | threshold fragility was localized; refined negative pair separated cleanly; ambiguity concentrated on $I_{\mathrm{ri}}$ handling and near-miss probe quality | positive stronger pair remained ambiguity-bound; near-identity probe still entangled | localized rather than promoted ‘P5-02‘, ‘P5-03‘, ‘P5-04‘ |
| Mini-resolution | can probe quality and $I_{\mathrm{ri}}$ handling be cleaned up without doctrinal weakening? | normalized-$I_{\mathrm{ri}}$ handling provisionally stabilized the harder positive pair; near-identity v2 clarified the probe weakness | near-identity v2 remained too weak to serve as a clean boundary probe | strengthened ‘P5-02‘, ‘P5-06‘; kept ‘P5-01‘, ‘P5-04‘ ambiguous |
| Residual resolution | can the open residuals be improved without doctrinal loosening? | ‘near_identity_v3‘ produced PASS/AMBIGUOUS/FAIL with $I_{\mathrm{ri}}$ first-fail; refined substrate families localized the live ambiguity to $I_{\mathrm{ri}}$ handling | the results were still only localized and internal | promoted both residual families to stronger localized support |
| High-value confirmation and Residual B hardening | do those gains survive a more independent judge path, a second family, and externalization-lite? | substrate-equivalence converged across two pair families; existence/rupture converged across two probe families under judge-v3 and pseudo-external reproduction | all support remained internal-only and a localized legacy caveat survived for ‘P5-02‘/‘P5-06‘ | promoted ‘P5-01‘, ‘P5-02‘, ‘P5-04‘, ‘P5-06‘ to robust support with localized caveats |

# 14 Priority Next Discriminators

The current state of the program already suggests which next moves are scientifically efficient. The relevant issue is not breadth but leverage: each next discriminator should either upgrade a provisional claim or localize an ambiguity into an implementation-limited corner.

| Priority target | Why it matters now | What the next narrow result would change | Linked claims |
|---|---|---|---|
| Independent external-style rerun of the two boundary families | current rupture/existence support is now robust internally but still lacks outside-program confirmation | an outside reproduction would test whether the family convergence is program-specific or genuinely portable | ‘P5-01‘, ‘P5-04‘ |
| Independent external-style rerun of both substrate families | current strongest positive equivalence path is robust internally but still carries a localized legacy $I_{\mathrm{ri}}$ caveat | an outside reproduction would test whether the strengthened equivalence path survives beyond the current judge lineage | ‘P5-02‘, ‘P5-06‘ |
| Matched external-style complexity baselines | the complexity-only result is internally strong but still lacks cross-program baseline pressure | a matched baseline campaign would turn the strongest present internal discriminator into a more externally credible one | ‘P5-05‘ |
| Independent maintained judge implementation | current hardening is substantial but still same-program | a separately maintained judge would reduce same-lineage measurement dependence without touching doctrine | all operational readings |

# 15 Residual Technical Caveats and Non-Claims

## 15.1 Residual Technical Caveats

The remaining caveats are narrow enough to state cleanly:

1. The strongest positive substrate-equivalence path is still partly dependent on normalized $I_{\mathrm{ri}}$ handling. This is a `METRIC_OR_TOLERANCE_LIMIT`, not a robust positive closure.
2. The current ‘near_identity_v2‘ probe remains an `IMPLEMENTATION_LIMIT`. It is cleaner than the original generator, but in the tested band it under-stresses the intended boundary.
3. Threshold stress is not globally problematic, but it is not perfectly flat either. The current result is localized rather than globally robust.
4. All empirical support remains `INTERNAL_SUPPORT_ONLY`. No external lab, external evaluator, or external benchmark suite has confirmed the framework. The clean-clone/fresh-environment re-instantiation strengthens same-machine release reproducibility only; it is not external validation, and no separate clean-agent confirmation artifact is being counted here as stronger evidence than that.

## 16 Final Claim Freeze Table

The present paper is a prediction-and-falsation layer, not a moving target. The table below records the narrowest honest freeze state for the main *Operational Consciousness Criterion* claim families at the end of the current phase.

| Claim | Status | Current blocker | Next step |
| --- | --- | --- | --- |
| ‘P5-01‘ | `ROBUST SUPPORT LOCALIZED CAVEAT` | support is now strong internally but still nearest to the legibility margin on the positive edge | rerun the convergent probe families under an evaluator maintained outside the current repo/runtime |
| ‘P5-02‘ | `ROBUST SUPPORT LOCALIZED CAVEAT` | one legacy raw continuous/discrete positive pair remains ambiguous on $I_{\mathrm{ri}}$ handling | reproduce both strengthened substrate families under a more independent evaluator without post hoc threshold loosening |
| ‘P5-03‘ | `PROVISIONAL SUPPORT LOCALIZED` | one localized knife-edge threshold band remains active in a negative-control region | rerun the localized stress zone with denser threshold sampling and fixed judge contract |
| ‘P5-04‘ | `ROBUST SUPPORT LOCALIZED CAVEAT` | rupture-side support is strong internally but still bounded to the present invariant package and frozen thresholds | rerun the same boundary families under a separately maintained evaluator and frozen release package |
| ‘P5-05‘ | `ROBUST INTERNAL SUPPORT` | no external matched-baseline campaign exists yet | add matched external-style complexity/activity baselines under the same judge contract |
| ‘P5-06‘ | `ROBUST SUPPORT LOCALIZED CAVEAT` | inherits the same localized substrate-equivalence caveat as ‘P5-02‘ | upgrade only if the strengthened substrate families survive a more independent external-style rerun |

## 17 Conclusion

The canon now has a prediction layer and a failure-mode ledger with explicit discriminators, explicit weakening conditions, and explicit support classes.

The current internal picture is sharp enough to summarize without inflation. The complexity/connectivity insufficiency claim has robust internal support. Substrate-preserved class membership and biological non-primacy now have robust support with localized caveats. Approximate stability is provisionally supported with a localized threshold caveat. Boundary-sensitive stress of existence and rupture now has robust support with localized caveats under convergent internal probe families and a more independent judge path. All of this nevertheless remains internal-only.

This remains an internal prediction-and-falsation ledger. It does not upgrade the frozen canon to external validation, theorem closure beyond the upstream papers, or claim-safe public release.

## References


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